s9_np — Model Card

Direct-velocity whole-body: reach a fixed target in *confondu*

RL_Scoutsan project

2026-09-24

1 Task and robot

A Scout V2 mobile base with a 6-DOF Doosan arm must bring its *gripper frame* coincident (*confondu* — position AND orientation) with a **fixed** target frame. This is the direct-velocity baseline (agent `ScoutsanV1_1b`): the policy commands base and joint velocities *directly*, in contrast to the WDLS/-Path B model that emits WDLS weights. This run (`s9_np`) is re-based onto the wb Path B physics so it can be compared to the WDLS models on matched dynamics. The target is a single fixed pose (no randomization); a whole-body-transplant warm start (padded ARM 14/6 → 14/8) seeds the arm skill while the grippy 4-wheel drive is learned fresh.

2 Observations (14), base_link frame

slice	symbol	meaning
[0:3]	gripper pos	gripper (mid-fingers + tip offset) position
[3:7]	gripper quat	gripper frame orientation (w-positive)
[7:10]	target pos	target position
[10:14]	target quat	target frame orientation (w-positive)

Errors: $d = \|p_G - p_T\|$ (reach), $\varphi = 2\arccos|\langle q_G, q_T\rangle|$ (geodesic orientation), baseD $= \|p_T\|$ in base frame (how far the base is from target).

3 Actions (8) — direct velocity

Six arm-joint velocity commands (deltas clamped to one `maxJointVelocityDeg` step per decision, cap 30 °/s) plus base v, ω (scaled by `maxLinearVel`=0.3, `maxAngularVel`=0.4). Base → wheels by differential drive (wheelRadius 0.16459, wheelSeparation 0.58306), each wheel's commanded velocity clamped to `maxWheelVelocityDeg`=150 °/s (wb-matched). The base drive + clamps are identical to the WDLS agents; only the *source* of v, ω differs (direct policy vs WDLS solve).

4 Reward and gate

Per step: a progress term r_1, orientation/position shaping, directional ($\pm$) terms that are *zeroed inside the gate* (deadbands, so stillness at the goal is optimal), and a rate penalty r_8; a one-time **+100** bonus on success. Exact constants/equations: the s9 stage report.

event	condition	reward
SUCCESS	$d <$ `successRadius` $= 0.03$ m **and** $\varphi <$ `orientToleranceRad` $= 0.05$ rad	+100, end
FAIL	collision/fall (early exit)	penalty, end
TIMEOUT	step cap reached	shaping only

5 Physics and model configuration

item	value
Algorithm	PPO (Stable-Baselines3 2.3.2), `MlpPolicy`
Warm start	padded ARM transplant 14/6 → 14/8 (arm kept, base rows zeroed)
base caps	`maxLinearVel` 0.3, `maxAngularVel` 0.4 (wb-matched)
wheel cap / drive	`maxWheelVelocityDeg` 150; force 100 / damping 5000
arm cap	`maxJointVelocityDeg` 30 °/s
obs / act	14 / 8 (continuous)
target	fixed (randomize position/orientation OFF)

6 Evaluation (deterministic, gate 0.03 m / 0.05 rad)

From `eval_s9np.py` (5 episodes, `deterministic=True`); per-step trace in `s9np_eval_steps.csv`, summary in `s9np_eval.json`, figures in `figures/`.

metric	value
Episodes	5
Success	5/5 = 100%
Mean min distance	0.026 m
Mean min orientation error	1.35° (0.0236 rad)
Decisions to reach the gate	≈26
End reasons (SUCCESS / FAIL / TIMEOUT)	5 / 0 / 0

The fixed target makes the run deterministic — the 5 trajectories overlap (a reproducibility check). Orientation reaches confondu ($< 2.9°$) by ∼step 13, then holds while the base finishes driving in; the base closes to ∼1.05 m and the arm does the fine reach.

Reproducibility

Model dir / prefix	`scoutsan_Baseline/models/model_scoutsan_s9_np` / `ppo_scoutsan_s9_np`
Scene	the s9 scene in the s9_np FIXED config (Behavior Default, 14/8, randomize OFF)
Eval / plots	`eval_s9np.py` → `eval_s9np_plots.py` → `build_s9np_figure_notes.py`
Video	3 cams (FollowCam/upcam/sidecam) + FrameAxesRenderer → `make_s9np_videos.sh`
Env	`unity-rl-gpu` (SB3 2.3.2, mlagents_envs 0.30.0), Editor port 5004
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